

a) The p.g.f. of $X + 1$ is

$$P_{X+1}(s) = \sum_{x=0}^{\infty} p_x s^{x+1} = s \sum_{x=0}^{\infty} p_x s^x = s P_X(s).$$

b) The p.g.f. of $3X$ is

$$P_{3X}(s) = \sum_{x=0}^{\infty} p_x s^{3x} = s \sum_{x=0}^{\infty} p_x (s^3)^x = s P_X(s^3).$$

c) Let $P(X \leq n) = q_n$ and $P(X = n) = p_n$.

Thus $q_n = P(X \leq n) = P(X = 0) + P(X = 1) + P(X = 2) + \cdots + P(X = n)$

$$= p_0 + p_1 + p_2 + \cdots + p_{n-1} + p_n = q_{n-1} + p_n$$

Thus, p.g.f. of q_n is

$$\begin{aligned} Q(s) &= \sum_{n=0}^{\infty} q_n s^n = \sum_{n=0}^{\infty} (q_{n-1} + p_n) s^n = \sum_{n=0}^{\infty} q_{n-1} s^n + \sum_{n=0}^{\infty} p_n s^n \\ &= s \sum_{n=1}^{\infty} q_{n-1} s^{n-1} + \sum_{n=0}^{\infty} p_n s^n \quad [\text{since } q_{-1} = 0] \\ &= s \sum_{n=1}^{\infty} q_{n-1} s^{n-1} + \sum_{n=0}^{\infty} p_n s^n \\ &= s Q(s) + P_X(s) \end{aligned}$$

which implies

$$Q(s)(1-s) = P_X(s).$$

Therefore,

$$Q(s) = \frac{P_X(s)}{1-s}$$

13.7 Chebyshev's Inequality

Let X be a random variable with mean μ and variance σ^2 . What is the probability that X is within t of its mean μ ? If we knew the exact distribution and probability function of X , then we could compute this probability: $P(|X - \mu| \leq t) = P(\mu - t \leq X \leq \mu + t)$.

There is another way to find a lower bound for this probability for almost any distribution that Chebychev's inequality can deliver. For instance, using Chebychev's Inequality we may obtain an expression like $P(|X - \mu| \leq 2) \geq 0.80$.

Theorem 13.7.1 Chebyshev's Inequality Let X be a random variable with mean μ and variance σ^2 . Then for any positive number $k (>1)$,

$$P(|X - \mu| > k\sigma) \leq \frac{1}{k^2} \quad \text{or} \quad P(|X - \mu| \leq k\sigma) \geq 1 - \frac{1}{k^2}.$$

Proof Let X be a continuous random variable with p.d.f. $f(x); -\infty < x < \infty$. By definition,

$$\sigma^2 = E(X - \mu)^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$

$$= \int_{-\infty}^{\mu-k\sigma} (x-\mu)^2 f(x) dx + \int_{\mu-k\sigma}^{\mu+k\sigma} (x-\mu)^2 f(x) dx + \int_{\mu+k\sigma}^{\infty} (x-\mu)^2 f(x) dx.$$

Since the second of the three terms of right side is non-negative,

$$\sigma^2 \geq \int_{-\infty}^{\mu-k\sigma} (x-\mu)^2 f(x) dx + \int_{\mu+k\sigma}^{\infty} (x-\mu)^2 f(x) dx. \quad (13.7.1)$$

From the first integral of (13.7.1), we have

$$x \leq \mu - k\sigma \Rightarrow \mu - x \geq k\sigma \Rightarrow (x - \mu)^2 \geq k^2 \sigma^2$$

and from the second of (13.7.1), we have

$$x \geq \mu + k\sigma \Rightarrow x - \mu \geq k\sigma \Rightarrow (x - \mu)^2 \geq k^2 \sigma^2.$$

It follows that

$$\sigma^2 \geq \int_{-\infty}^{\mu-k\sigma} k^2 \sigma^2 f(x) dx + \int_{\mu+k\sigma}^{\infty} k^2 \sigma^2 f(x) dx$$

which implies that

$$\int_{-\infty}^{\mu-k\sigma} f(x) dx + \int_{\mu+k\sigma}^{\infty} f(x) dx \leq \frac{1}{k^2}$$

Hence,

$$\int_{-\infty}^{\infty} f(x) dx - \int_{-\infty}^{\mu-k\sigma} f(x) dx - \int_{\mu+k\sigma}^{\infty} f(x) dx \geq 1 - \frac{1}{k^2}$$

which implies

$$P(\mu - k\sigma \leq X \leq \mu + k\sigma) \geq 1 - \frac{1}{k^2}$$

Or alternatively,

$$P(|X - \mu| \leq k\sigma) \geq 1 - \frac{1}{k^2}.$$

This proves the theorem.

Note: If $k\sigma = c$, then

$$P(|X - \mu| \leq c) \geq 1 - \frac{\sigma^2}{c^2}$$

$$\text{Or, } P(|X - \mu| > c) \leq \frac{\sigma^2}{c^2}.$$

To compute probability of an event described in terms of a random variable X , we need the p.d.f. of X . Chebyshev's inequality, which does not depend on the distribution of X , to bound the tail probabilities of X having its mean and finite variance.

Example 13.19 The voltage in a certain circuit is a random variable with mean 120 and deviation 5. Sensitive equipment will be damaged if the voltage is not between 112 and 128. Use Chebyshev's inequality to bound the probability of damage.

Solution Let X be the voltage of the circuit. The problem here is to find

$$P(X > 128 \text{ or } X < 112) = 1 - P(112 \leq X \leq 128).$$

By Chebyshev's theorem for any $k > 1$, we have

$$P(|X - \mu| \leq k\sigma) \geq 1 - \frac{1}{k^2}$$

For $\mu = 120$ and $\sigma = 5$,

$$1 - P(112 \leq X \leq 128)$$

$$= 1 - P(-8 \leq X - 120 \leq 8) = 1 - P(|X - 120| \leq 8) = P(|X - 120| > 8) \leq \frac{1}{k^2}.$$

Now, $k\sigma = 8$. Therefore, $k = 8/\sigma = 8/5$.

Thus, $P(|X - 120| > 8) \geq \frac{25}{64}$.

Example 13.20 Using Chebyshev's inequality, find how many times a fair coin must be tossed in order that the probability will be at least 0.90 that the ratio of the observed number of heads to the total number of tosses will lie between 0.4 and 0.6.

Solution Let X denote the observed number of heads in n tosses. Then

$$P\left(\left|\frac{X}{n} - p\right| < c\right) \geq 1 - \frac{\sigma^2}{c^2}.$$

Since the coin is unbiased, $p = 0.5$ and since the value of X/n will be between 0.4 and 0.6, we have

$$\left|\frac{X}{n} - p\right| \leq 0.1; \text{ let } c = 0.1.$$

Therefore, $P\left(\left|\frac{X}{n} - p\right| < 0.1\right) \geq 1 - \frac{pq}{n(0.1)^2}$ [since $\sigma^2 = V\left(\frac{X}{n}\right) = \frac{pq}{n}$]

$$\text{Now, } 1 - \frac{0.5 \times 0.5}{n \times 0.01} \geq 0.9 \Rightarrow \frac{0.25}{0.01 \times n} \leq 0.1$$

$$\text{which implies } n \geq \frac{0.25}{0.01 \times 0.1} = 250.$$

Hence the coin should be tossed at least 250 times.

Example 13.21 Given that the random variable X has the density function,

$$f(x) = \begin{cases} \frac{1}{2\sqrt{3}} & ; -\sqrt{3} < x < \sqrt{3}, \\ 0 & ; \text{otherwise.} \end{cases}$$

Compute $P\left(|X| \geq \frac{3}{2}\right)$ and $P(|X| \geq 2)$. Compare these probabilities with the upper bounds given by

Chebyshev's inequality.

Solution Chebyshev's inequality states that when a random variable X has finite variance σ^2 ,

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

We need mean μ and variance σ^2 of the distribution.

$$\mu = \int_{-\sqrt{3}}^{\sqrt{3}} x \frac{1}{2\sqrt{3}} dx = \frac{1}{2\sqrt{3}} \frac{x^2}{2} \Big|_{-\sqrt{3}}^{\sqrt{3}} = \frac{1}{4\sqrt{3}} (3 - 3) = 0,$$

and $\sigma^2 = E(X^2) - \mu^2$.

$$\begin{aligned} &= E(X^2) - 0 = E(X^2) \\ &= \int_{-\sqrt{3}}^{\sqrt{3}} x^2 \frac{1}{2\sqrt{3}} dx = \frac{1}{2\sqrt{3}} \frac{x^3}{3} \Big|_{-\sqrt{3}}^{\sqrt{3}} \\ &= \frac{1}{6\sqrt{3}} [(\sqrt{3})^3 - (-\sqrt{3})^3] = 1 \end{aligned}$$

Now, $P\left(|X| \geq \frac{3}{2}\right) = 1 - P\left(|X| \leq \frac{3}{2}\right)$

$$\begin{aligned} &= 1 - P\left(-\frac{3}{2} \leq X \leq \frac{3}{2}\right) \\ &= 1 - \int_{-3/2}^{3/2} \frac{1}{2\sqrt{3}} dx = 1 - \frac{1}{2\sqrt{3}} x \Big|_{-3/2}^{3/2} \\ &= 1 - \frac{1}{2\sqrt{3}} \frac{6}{2} = 1 - \frac{3}{2\sqrt{3}} = 1 - \frac{\sqrt{3}}{2} \approx 0.134. \end{aligned}$$

Using Chebyshev's inequality with $\mu = 0$ and $\sigma^2 = 1$, we obtain

$$P\left(|X| \geq \frac{3}{2}\right) \leq \frac{4}{9} \approx 0.444.$$

That is, true probability 0.134 is much less than the upper bound given by Chebyshev's inequality.

Next $P(|X| \geq 2) = 1 - P(|X| \leq 2)$

$$= 1 - P(-2 \leq X \leq 2) = 1 - 1 = 0 \text{ [because, } -2 < -\sqrt{3} \leq X \leq \sqrt{3} < 2].$$

Using Chebyshev's inequality to find the upper bound with $\mu = 0$ and $\sigma^2 = 1$, we have

$$P(|X - 0| \geq 2.1) \leq \frac{1}{(2)^2} = \frac{1}{4} = 0.25.$$

Again, 0 is much less than the upper bound 0.25.

Key Terms

Raw moments: r th raw moment of a random variable X is $E(X^r)$

Central moments: r th central moment of a random variable X is $E[\{X - E(X)\}^r]$

Skewness: One measure of skewness is $\frac{E[(X - \mu)^3]}{\sigma^3}$